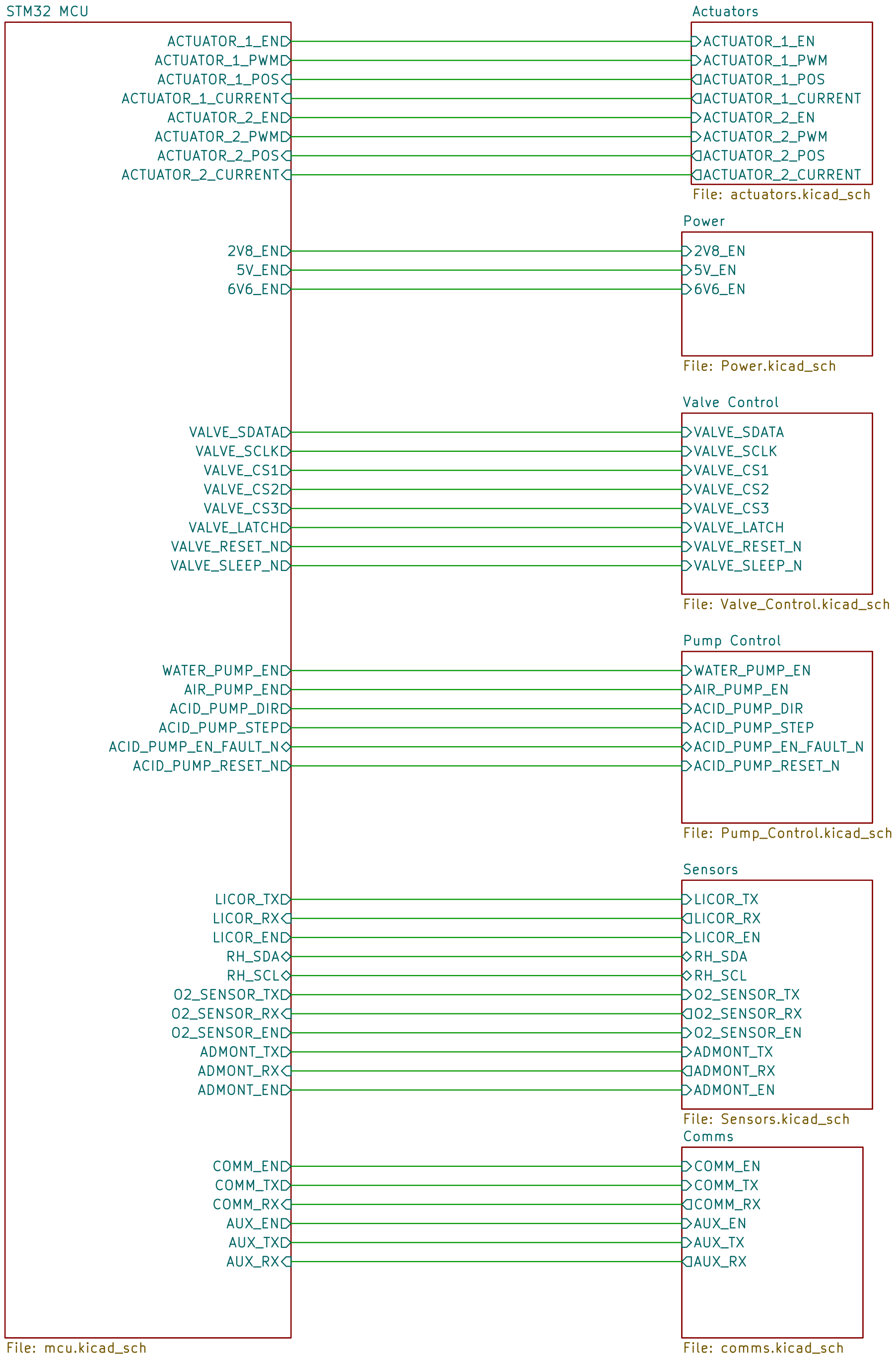

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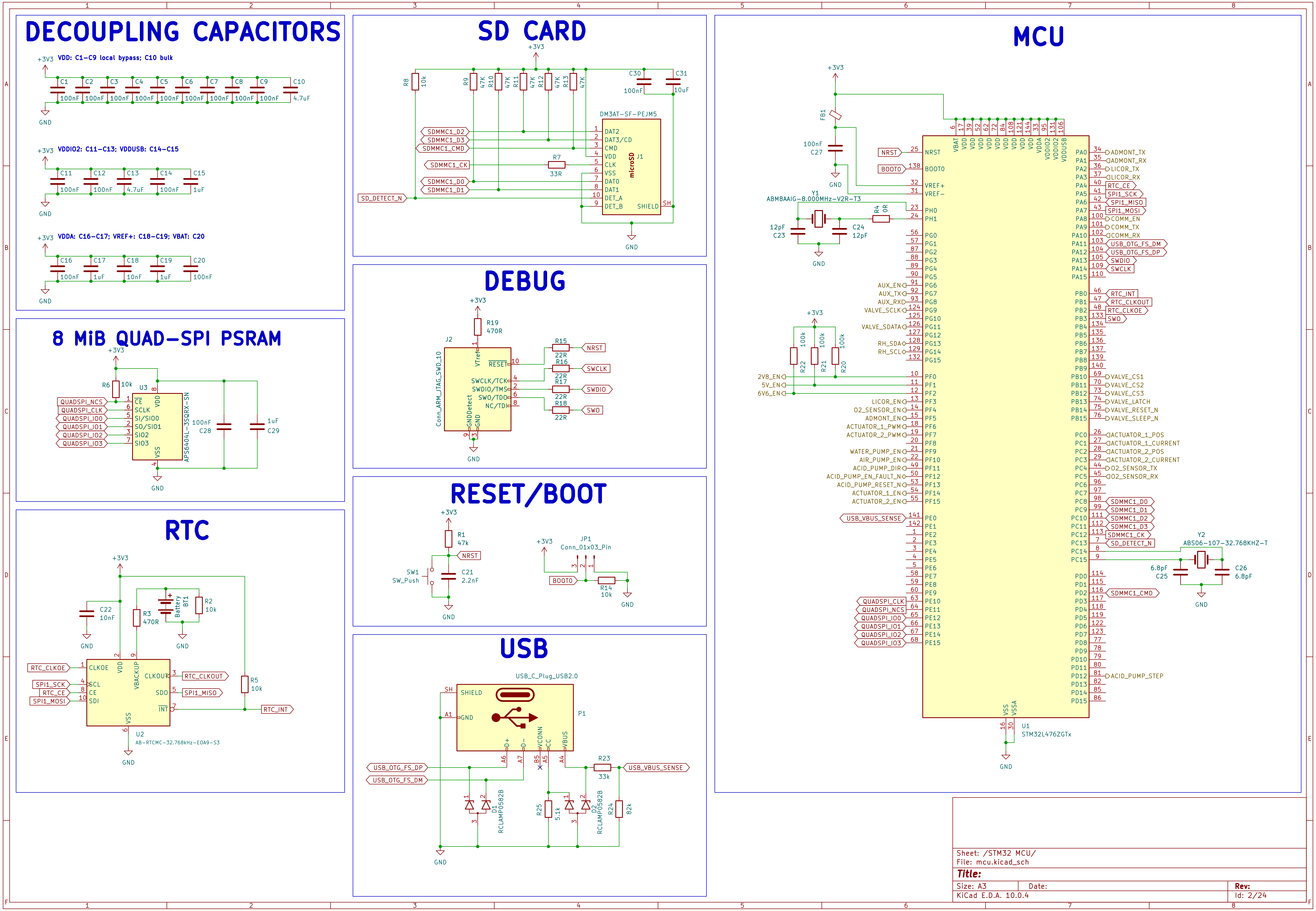
Title:

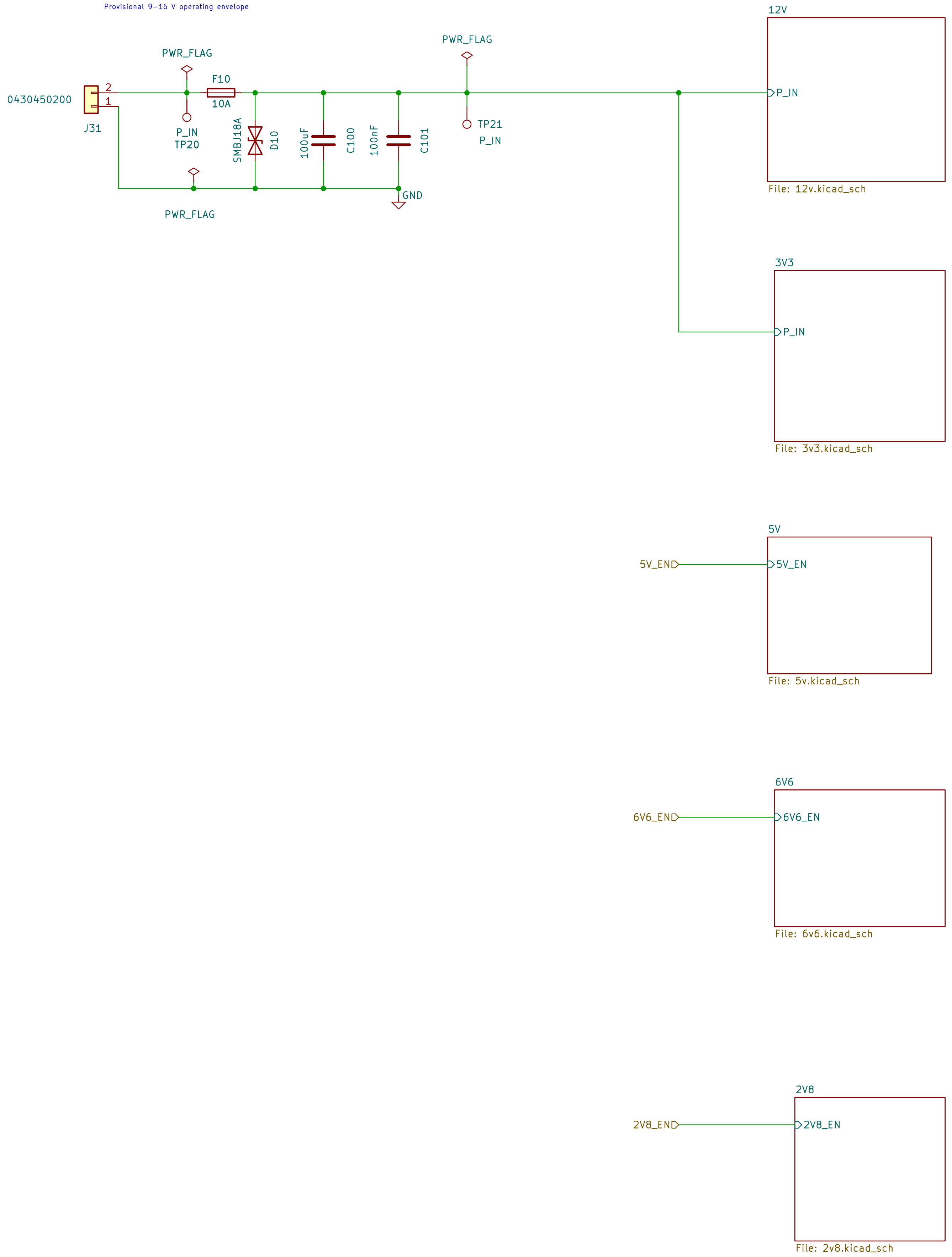
Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 1/24


DECOUPLING CAPACITORS
+3V3 VDD: C1–C9 local bypass; C10 bulk
C1 C2 C3 C4 C5 C6 C7 C8 C9 C10
100nF 100nF 100nF 100nF 100nF 100nF 100nF 100nF 100nF 4.7uF
GND
+3V3 VDDIO2: C11–C13; VDDUSB: C14–C15
C11 100nF C12 100nF C13 4.7uF C14 100nF C15 1uF
+3V3 VDDA: C16–C17; VREF+: C18–C19; VBAT: C20
C16 100nF C17 1uF C18 10nF C19 1uF C20 100nF
8 MiB QUAD–SPI PSRAM
+3V3
R6 10k
U3
QUADSPI_NCS QUADSPI_CLK QUADSPI_IO0 QUADSPI_IO1 QUADSPI_IO2 QUADSPI_IO3
CE SCLK SI/SIO0 SO/SIO1 SIO2 SIO3 VDD VSS
APS6404L–3SQRX–SN
100nF C28
1uF C29
GND
RTC
+3V3
C22 10nF
R3 470R
BT1 Battery
R2 10k
R5 10k
RTC_CLKOE SPI1_SCK RTC_CE SPI1_MOSI
CLKOE SCL CE SDI VDD VBACKUP CLKOUT SDO INT VSS
RTC_CLKOUT SPI1_MISO RTC_INT
U2
AB–RTCMC–32.768kHz–EOA9–S3
GND
SD CARD
+3V3
R8 10k R9 47K R10 47K R11 47K R12 47K R13 47K
C30 100nF C31 10uF
DM3AT–SF–PEJM5
SDMMC1_D2 SDMMC1_D3 SDMMC1_CMD SDMMC1_CK SDMMC1_D0 SDMMC1_D1 SD_DETECT_N
R7 33R
DAT2 DAT3/CD CMD VDD CLK VSS DAT0 DAT1 DET_A DET_B SHIELD
microSD J1
GND
DEBUG
+3V3
R19 470R
J2
Conn_ARM_JTAG_SWD_10
VTref RESET SWCLK/TCK SWDIO/TMS SWO/TDO NC/TDI GNDDetect GND
R15 22R R16 22R R17 22R R18 22R
NRST SWCLK SWDIO SWO
GND
RESET/BOOT
+3V3
R1 47k
NRST
SW1 SW_Push
C21 2.2nF
GND
+3V3
JP1 Conn_01x03_Pin
BOOT0
R14 10k
GND
USB
USB_C_Plug_USB2.0
SHIELD GND D+ D– VCONN CC VBUS
P1
USB_OTG_FS_DP USB_OTG_FS_DM
R23 33k
USB_VBUS_SENSE
D1 RCLAMP0582B
R25 5.1k
D2 RCLAMP0582B
R24 82k
GND
MCU
+3V3
FB1
100nF C27
NRST BOOT0
VREF+ VREF–
Y1 ABMBAAIG–8.000MHz–V2R–T3
R4 0R
12pF C23 C24 12pF
GND
AUX_EN AUX_TX AUX_RXD VALVE_SCLK VALVE_SDATA RH_SDA RH_SCL
+3V3
R22 100k R21 100k R20 100k
2V8_EN 5V_EN 6V6_EN
LICOR_EN O2_SENSOR_EN ADMONT_EN ACTUATOR_1_PWM ACTUATOR_2_PWM WATER_PUMP_EN AIR_PUMP_EN ACID_PUMP_DIR ACID_PUMP_EN_FAULT_N ACID_PUMP_RESET_N ACTUATOR_1_EN ACTUATOR_2_EN
USB_VBUS_SENSE
QUADSPI_CLK QUADSPI_NCS QUADSPI_IO0 QUADSPI_IO1 QUADSPI_IO2 QUADSPI_IO3
ADMONT_TX ADMONT_RX LICOR_TX LICOR_RX RTC_CE SPI1_SCK SPI1_MISO SPI1_MOSI COMM_EN COMM_TX COMM_RX USB_OTG_FS_DM USB_OTG_FS_DP SWDIO SWCLK
RTC_INT RTC_CLKOUT RTC_CLKOE SWO
VALVE_CS1 VALVE_CS2 VALVE_CS3 VALVE_LATCH VALVE_RESET_N VALVE_SLEEP_N
ACTUATOR_1_POS ACTUATOR_1_CURRENT ACTUATOR_2_POS ACTUATOR_2_CURRENT O2_SENSOR_TX O2_SENSOR_RX
SDMMC1_D0 SDMMC1_D1 SDMMC1_D2 SDMMC1_D3 SDMMC1_CK SD_DETECT_N
Y2 ABS06–107–32.768KHZ–T
6.8pF C25 C26 6.8pF
SDMMC1_CMD
ACID_PUMP_STEP
U1
STM32L476ZGTx
GND
Sheet: /STM32 MCU/
File: mcu.kicad_sch
Title:
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 2/24

POWER ENTRY

Provisional 9–16 V operating envelope



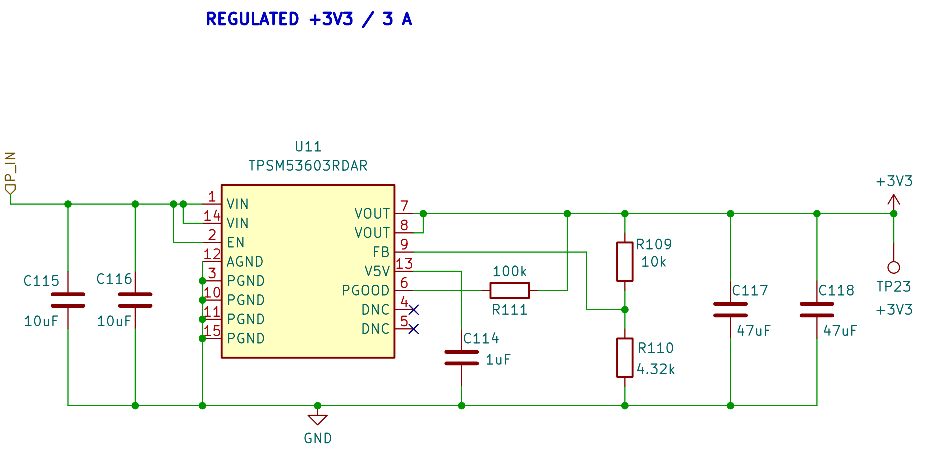
DESIGN NOTES

– The 12 V converter is provisionally designed for 6 A continuous and actuator inrush transients.

– Check the U10 RPM0026A and Q100/Q101 TO–263–3 land patterns before PCB placement.

– Input fuse/TVS and reverse–polarity protection remain provisional until the source is frozen.

Sheet: /Power/		
File: Power.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/24

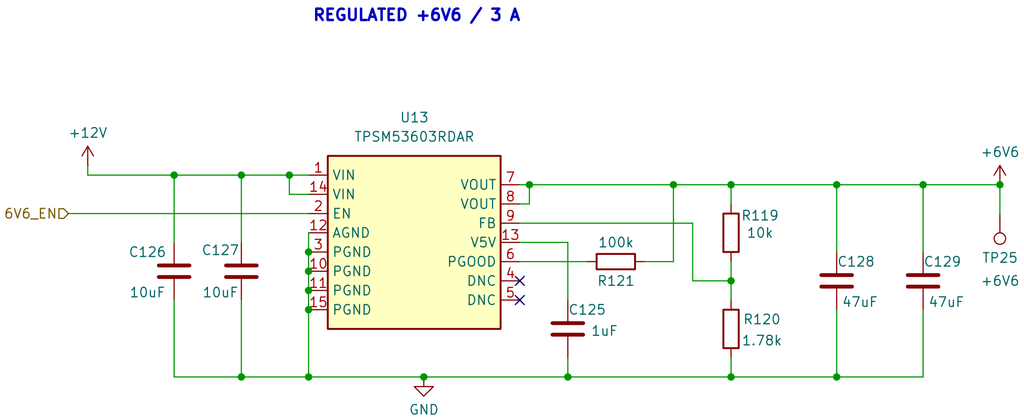


Sheet: /Power/3V3/
File: 3v3.kicad_sch

Title:

Size: A4 Date: **Rev:**

KiCad E.D.A. 10.0.4 Id: 4/24



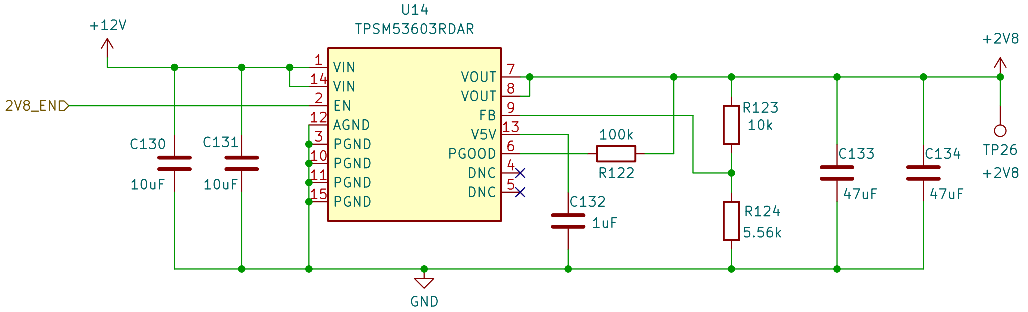
Sheet: /Power/6V6/
File: 6v6.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 5/24

REGULATED +2V8 / 3 A

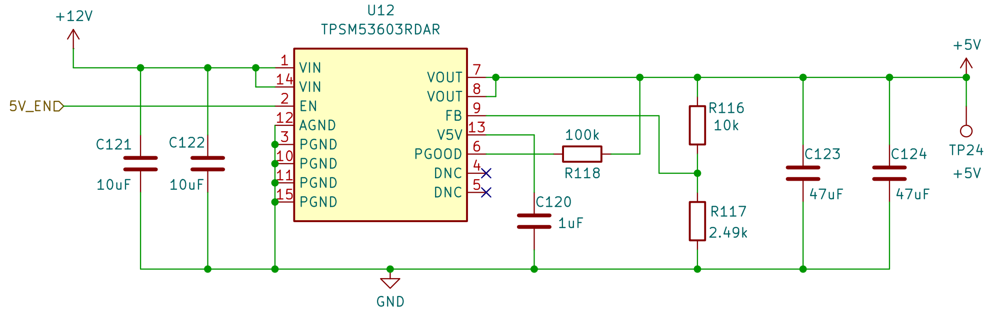


Sheet: /Power/2V8/
File: 2v8.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/24

REGULATED +5V / 3 A



Sheet: /Power/5V/
File: 5v.kicad_sch

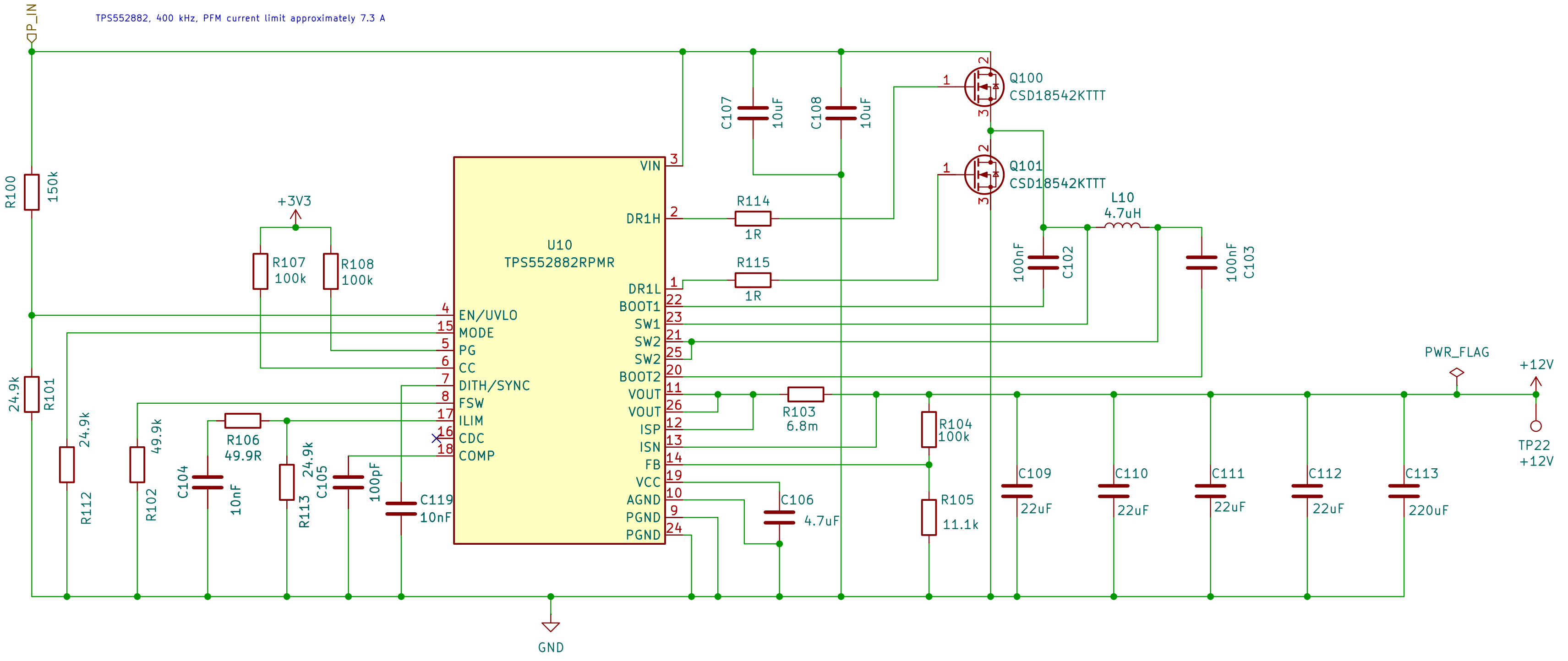
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 7/24

REGULATED +12 V / 6 A BUCK-BOOST

TPS552882, 400 kHz, PFM current limit approximately 7.3 A

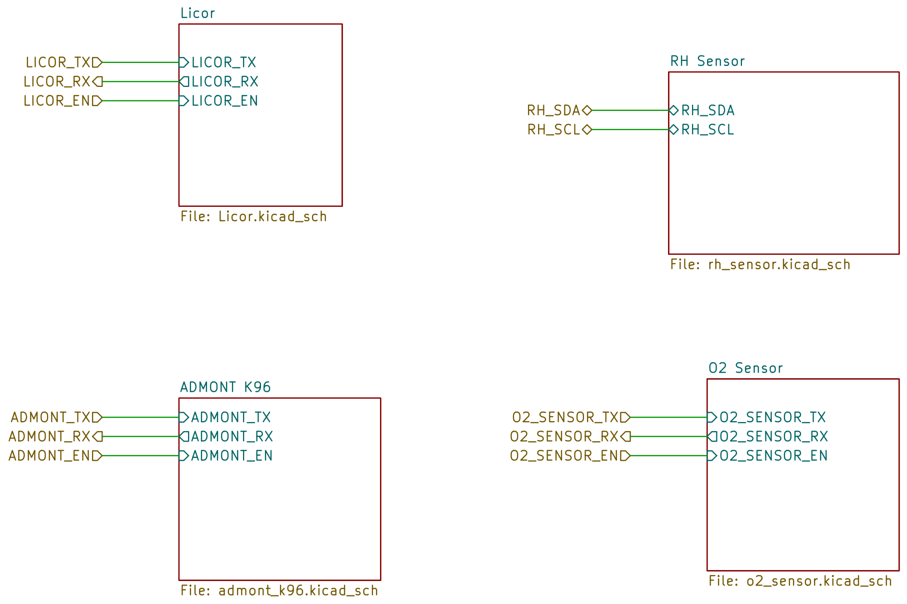


Sheet: /Power/12V/

File: 12v.kicad_sch

Title:

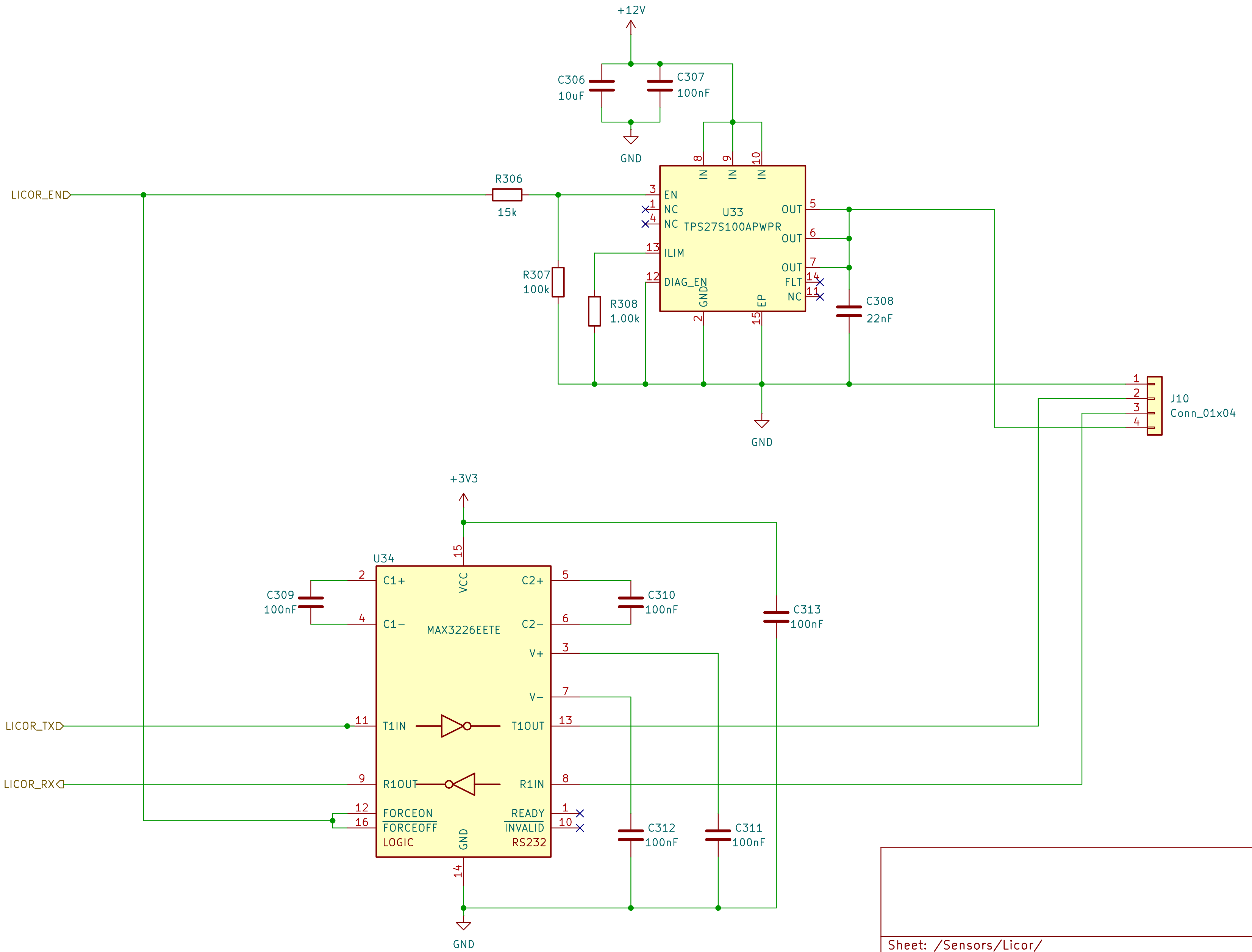
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 8/24



Sheet: /Sensors/
File: Sensors.kicad_sch

Title: Sensors

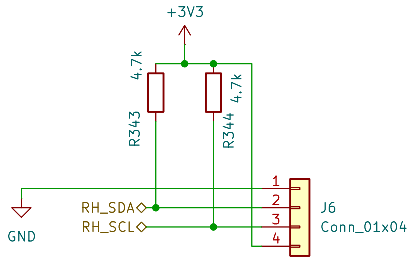
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 9/24



Sheet: /Sensors/Licor/
File: Licor.kicad_sch

Title:

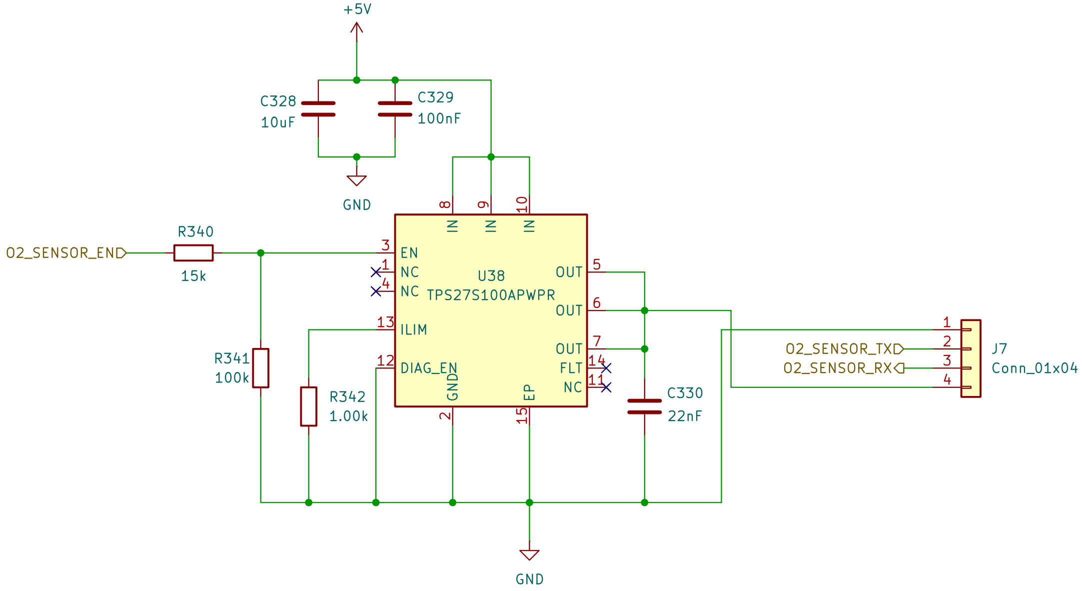
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 10/24



Sheet: /Sensors/RH Sensor/
File: rh_sensor.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 11/24

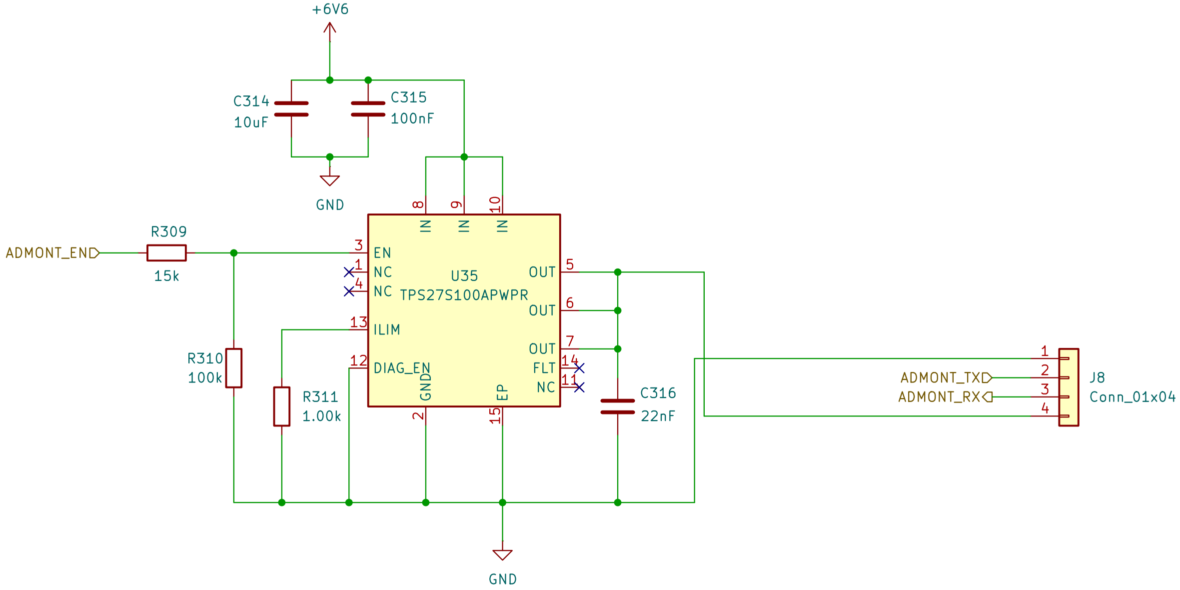


Sheet: /Sensors/O2 Sensor/
File: o2_sensor.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 12/24



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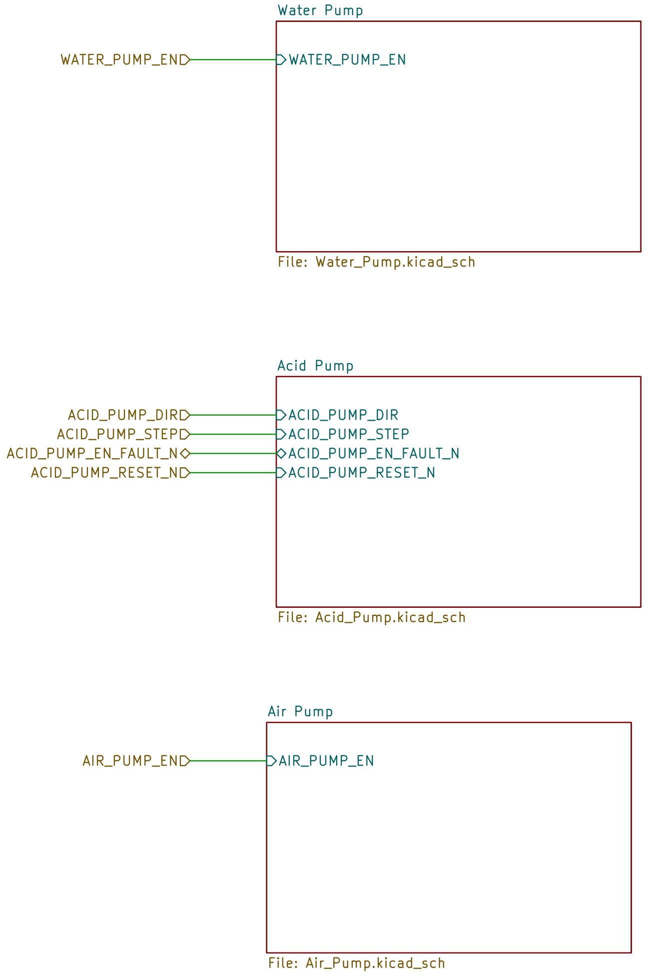
Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 13/24



Sheet: /Pump Control/
File: Pump_Control.kicad_sch

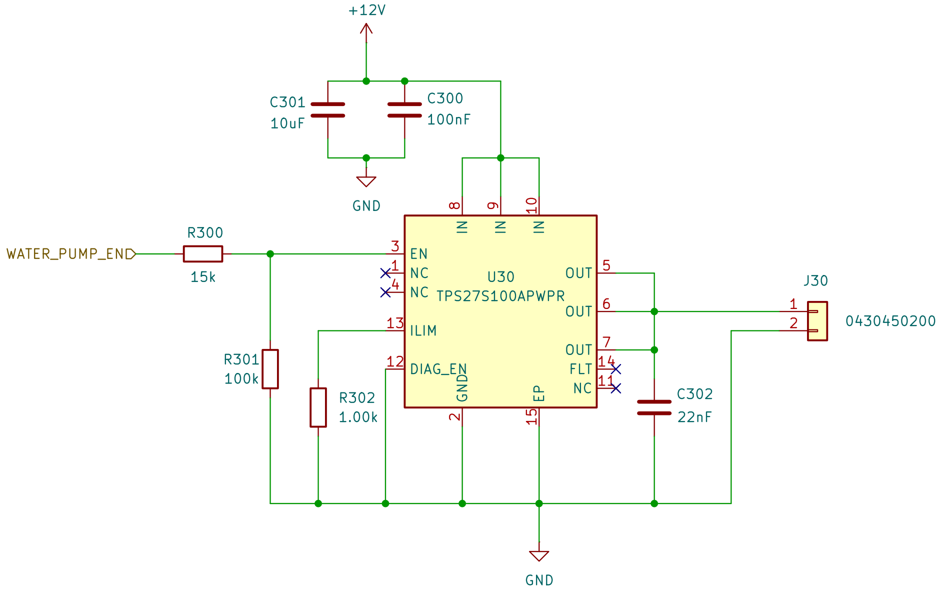
Title: Pump Control

Size: A4 Date: **Rev:**
KiCad E.D.A. 10.0.4 Id: 14/24

SBE 5P: 10–18 V option 1.8 A maximum turn–on surge, approximately 0.25 s.

TPS27S100A: 40 V / 4 A switch R302 = 1.00 kOhm sets approximately 2.5 A limit.

Pump has internal reverse–polarity diode TPS27S100A provides inductive turn–off clamp.



TPS27S100A smart switch; nominal 2.5–A current limit
Sea–Bird SBE 5P pump, 12–V high–side control

Sheet: /Pump Control/Water Pump/
File: Water_Pump.kicad_sch

Title: Water Pump

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 15/24

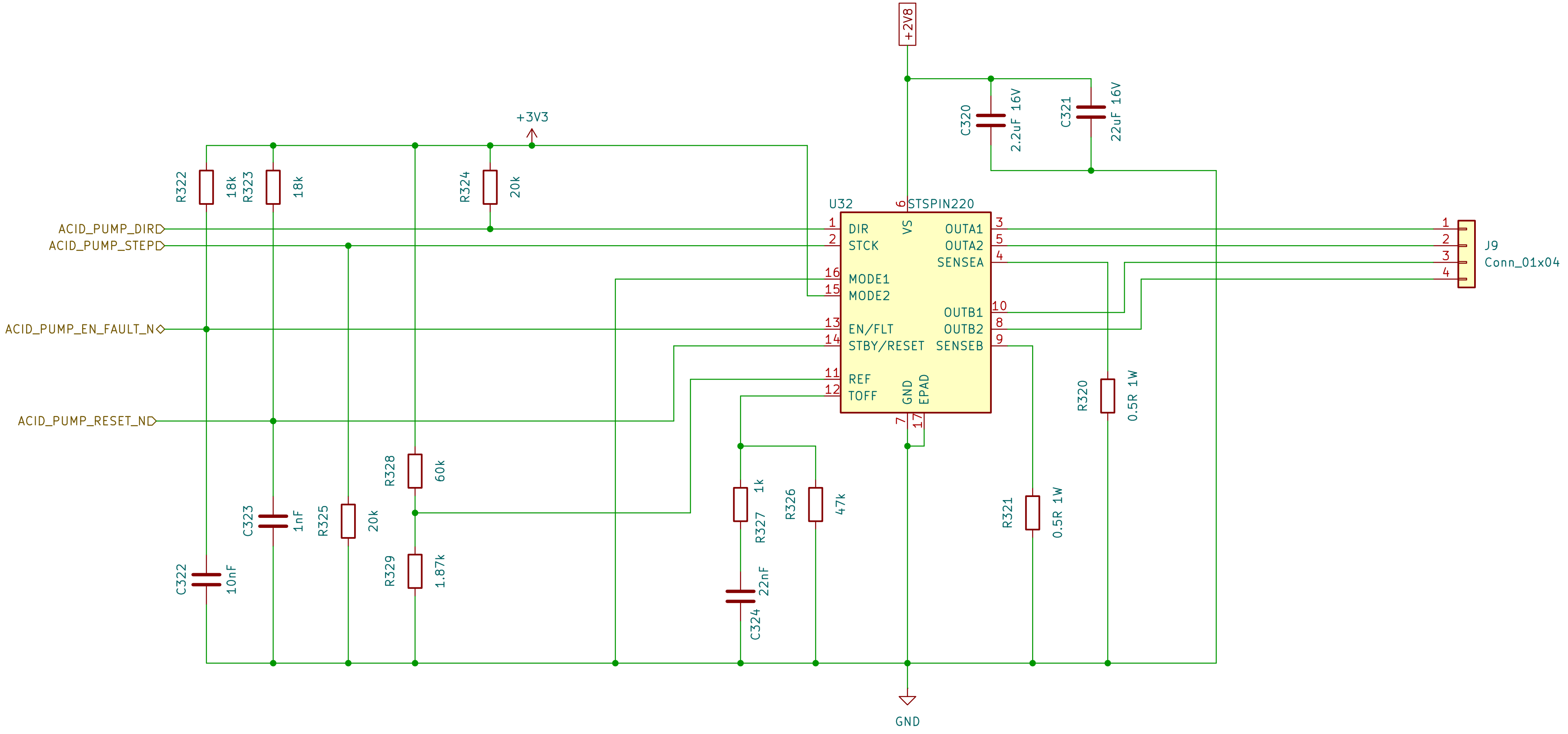
ACID PUMP STEPPER DRIVER STSPIN220, +2V8 MOTOR SUPPLY

Fixed 1/4-step latch: MODE1=0, MODE2=1, STEP=0 and DIR=1 when STBY/RESET rises.

Firmware must pull STEP low and DIR high while releasing RESET afterwards they are normal controls.

Current limit: 0.5 ohm sense, VREF adjustable 0-0.471 V I_PEAK approximately VREF/RSENSE = 0-0.94 A.

Place C320/C321 and both sense resistors close to U32. EPAD requires solid GND and thermal vias.

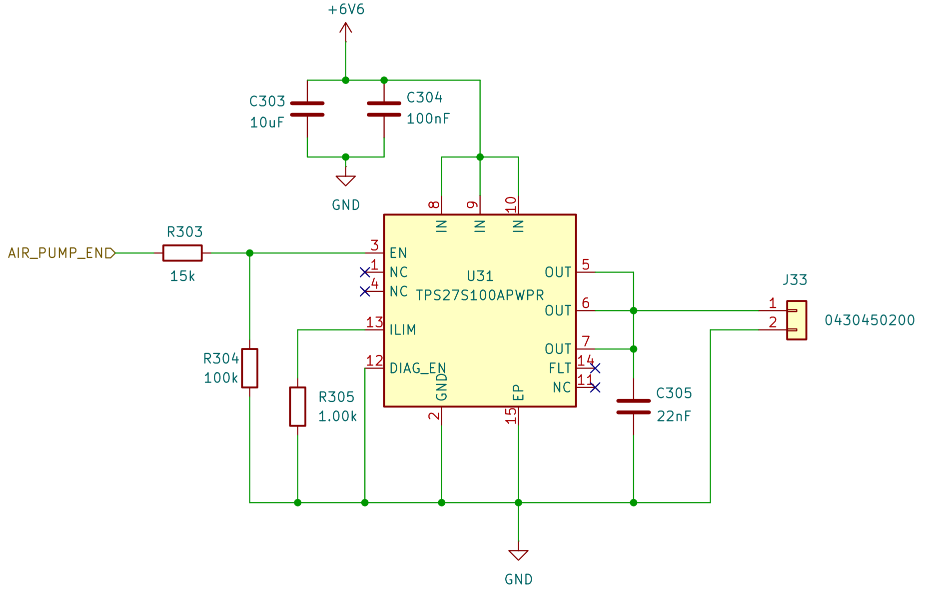


+2V8 motor supply; adjustable 0-0.94 A peak current limit
STSPIN220 bipolar stepper driver; fixed 1/4 microstep

Sheet: /Pump Control/Acid Pump/
File: Acid_Pump.kicad_sch

Title: Acid Pump

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 16/24

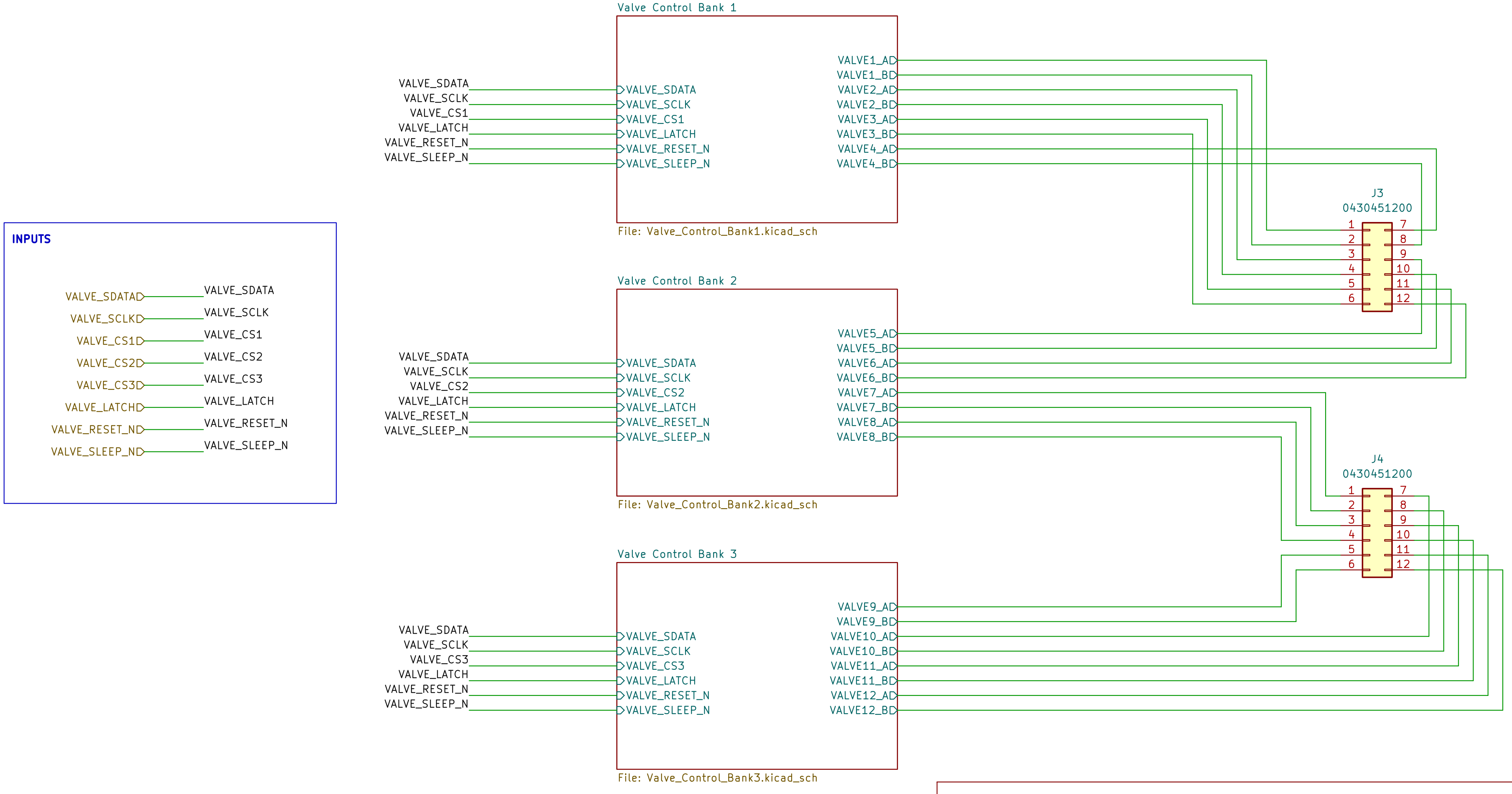


Sheet: /Pump Control/Air Pump/
File: Air_Pump.kicad_sch

Title: Air Pump

Size: A4 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 17/24

VALVE CONTROL — THREE CHILD BANKS, 12 POLARITY-REVERSING CHANNELS

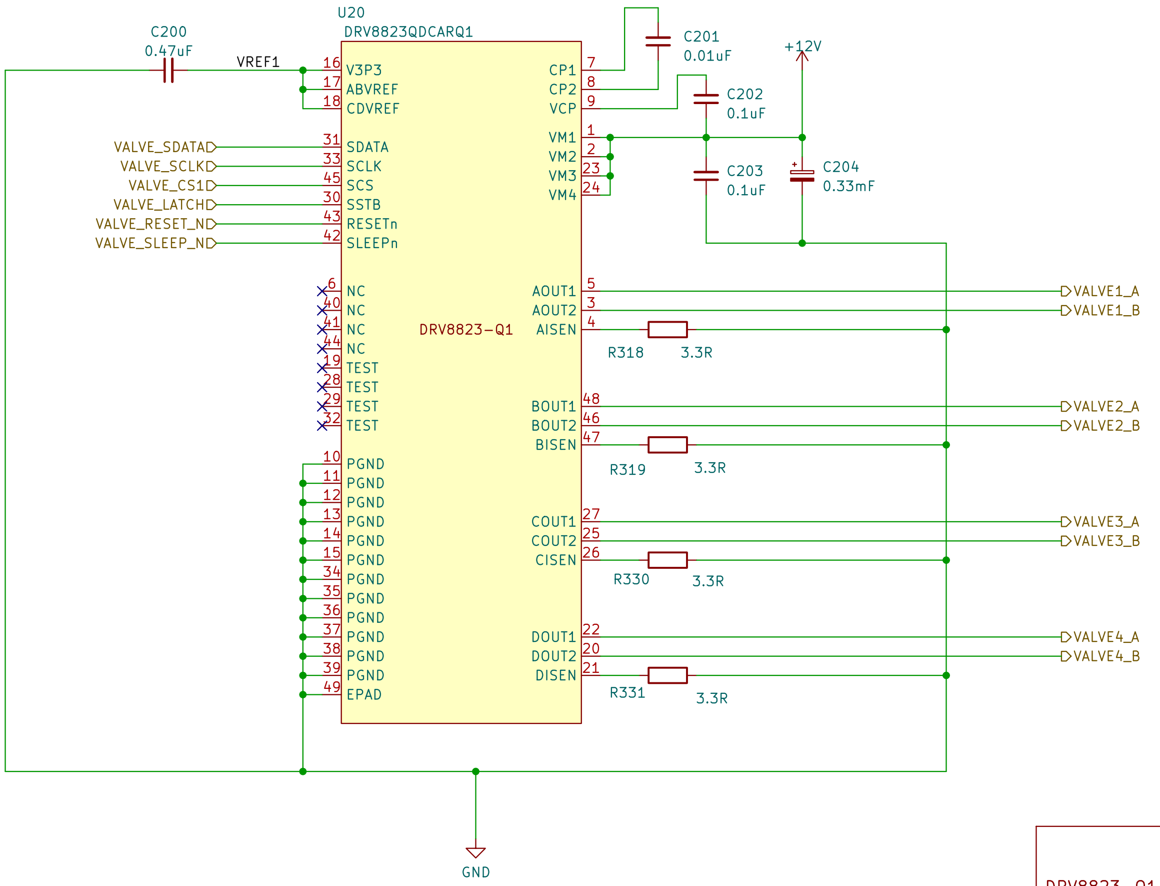


Channels 11–12 are reserved spares in the 10-valve population
Three DRV8823-Q1 child banks; 12 H-bridges for 4/8/10-valve variants

Sheet: /Valve Control/
File: Valve_Control.kicad_sch

Title: Valve Control

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 18/24



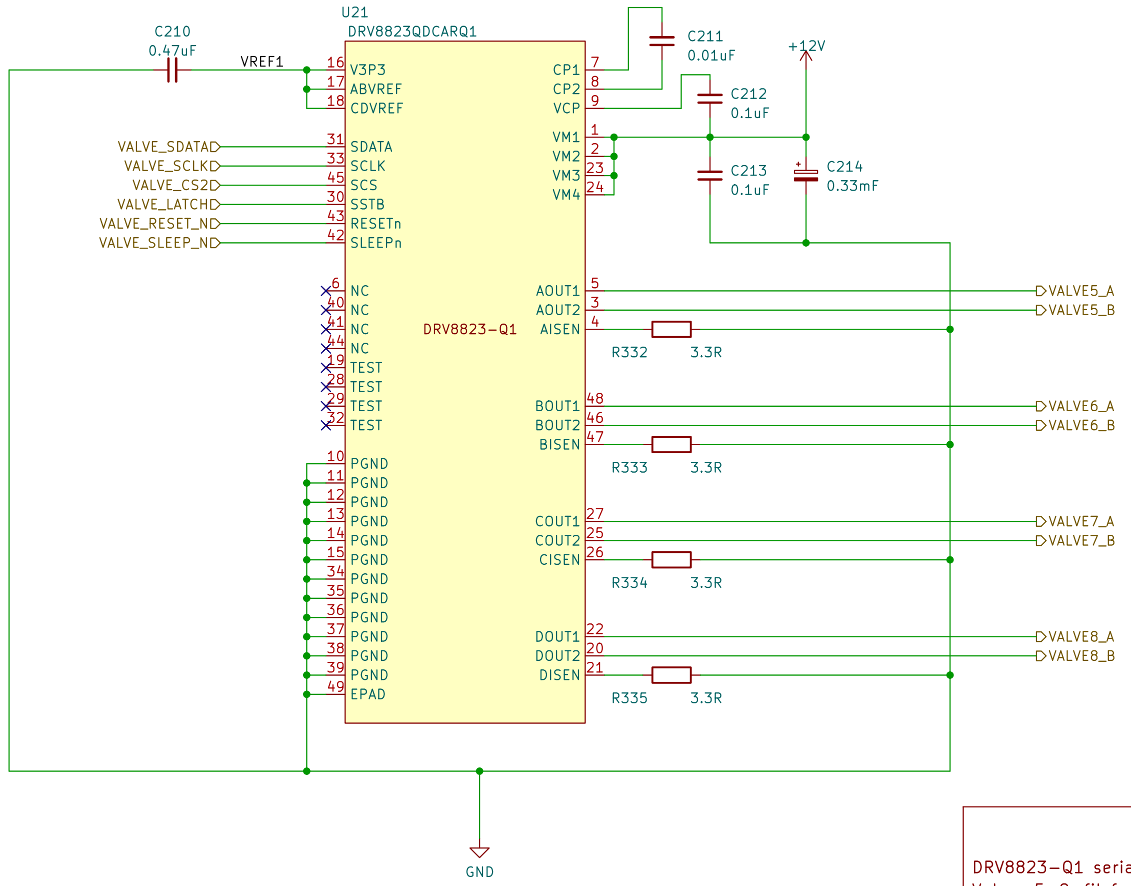
DRV8823-Q1 serial quad H-bridge
Valves 1-4; fit all 4/8/10-valve variants

Sheet: /Valve Control/Valve Control Bank 1/
File: Valve_Control_Bank1.kicad_sch

Title: Valve Control Bank 1

Size: A4 | Date: | Rev:

KiCad E.D.A. 10.0.4 | Id: 19/24



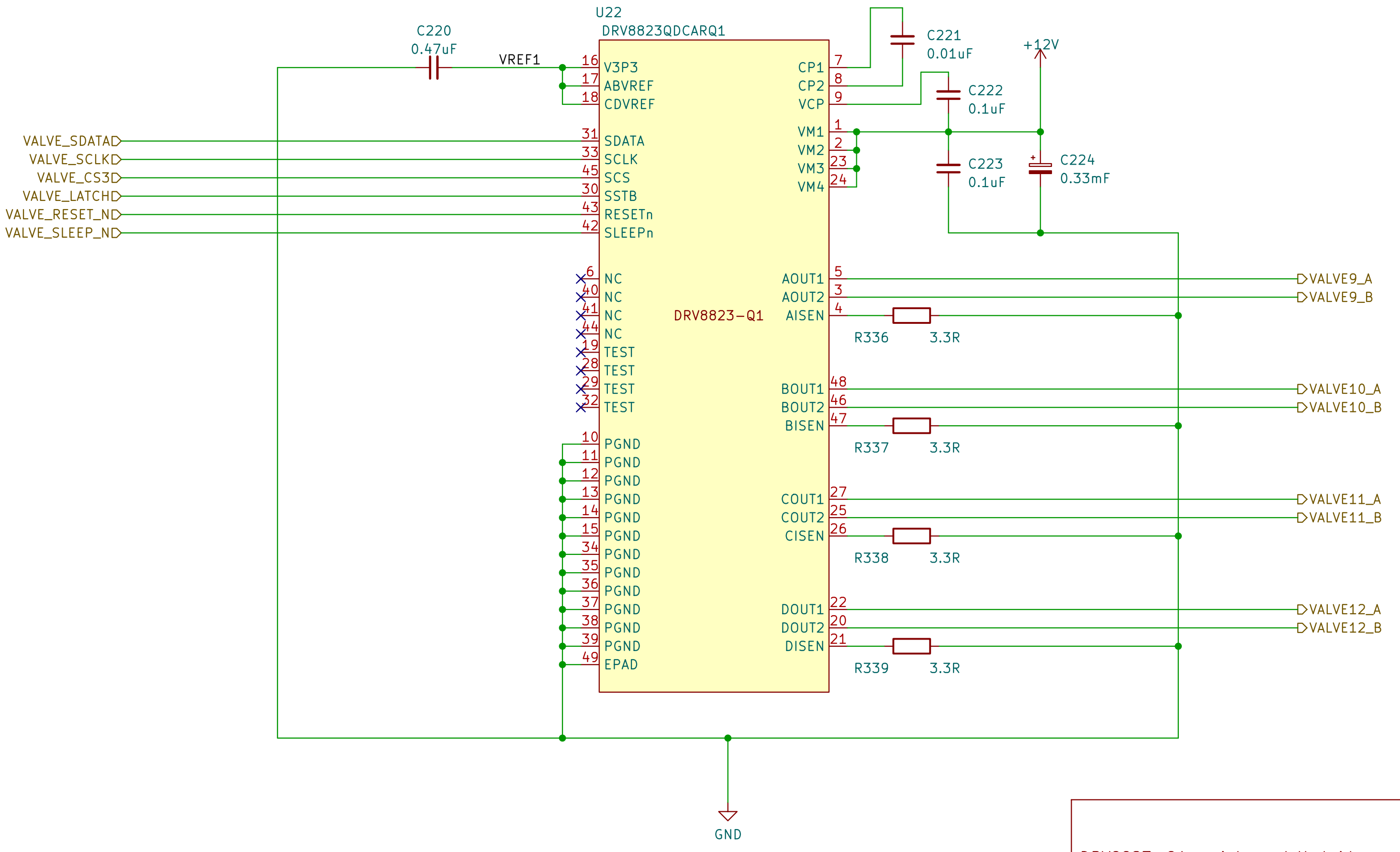
DRV8823-Q1 serial quad H-bridge
Valves 5-8; fit for 8/10-valve variants

Sheet: /Valve Control/Valve Control Bank 2/
File: Valve_Control_Bank2.kicad_sch

Title: Valve Control Bank 2

Size: A4 | Date: | Rev:

KiCad E.D.A. 10.0.4 | Id: 20/24

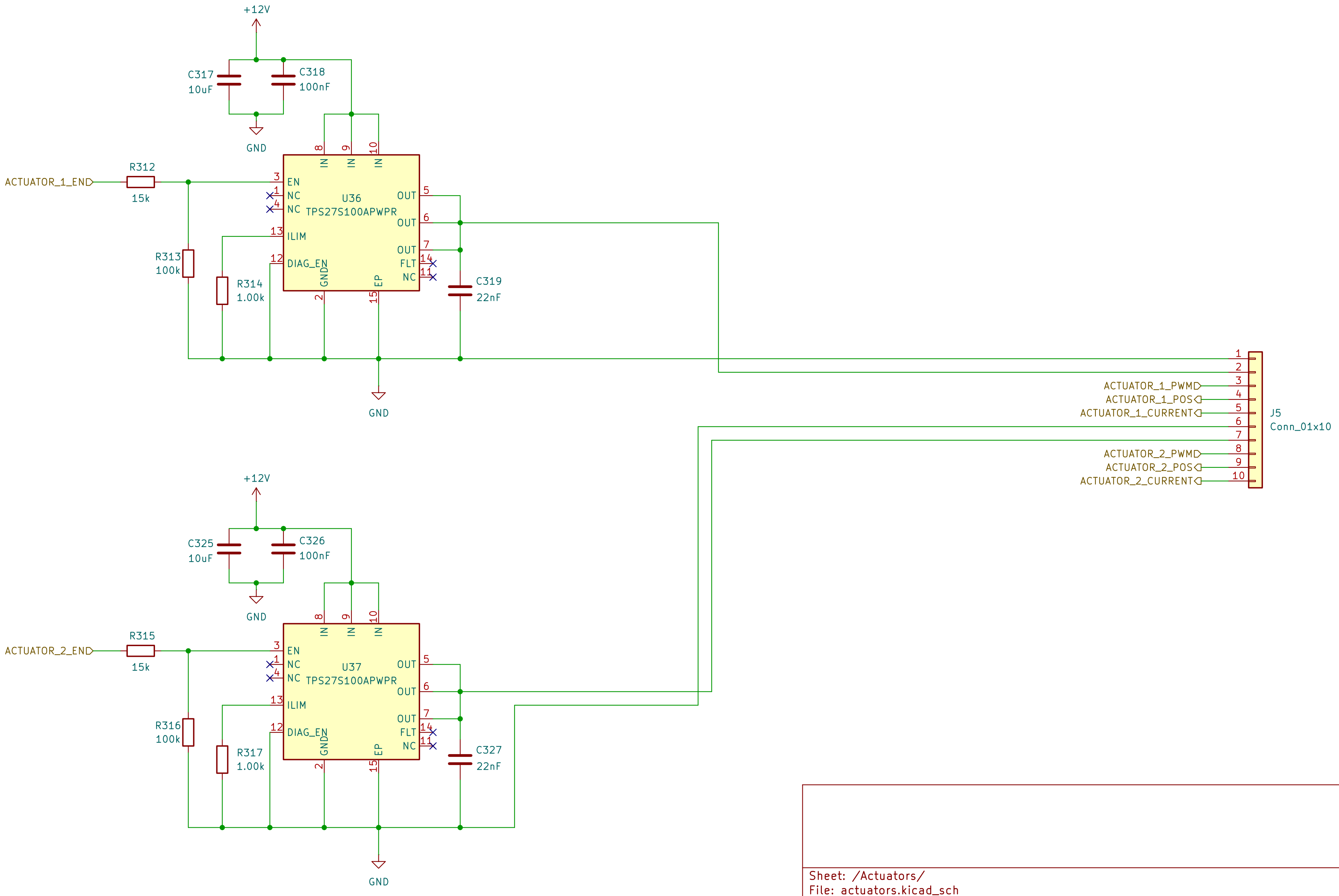


DRV8823-Q1 serial quad H-bridge
Valves 9-10 plus two spare channels; fit for 10-valve variant

Sheet: /Valve Control/Valve Control Bank 3/
File: Valve_Control_Bank3.kicad_sch

Title: Valve Control Bank 3

Size: A4 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 21/24



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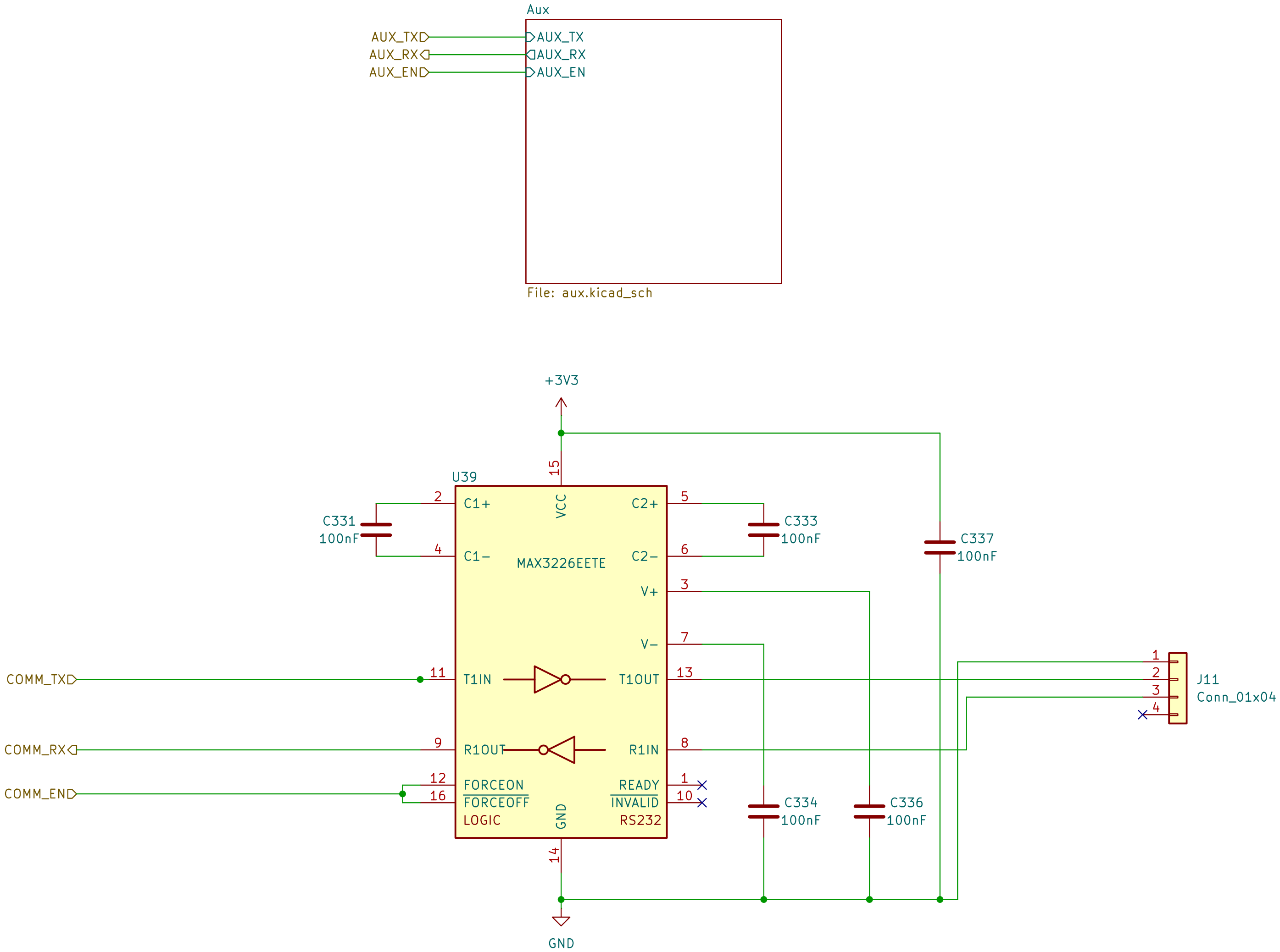
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 22/24



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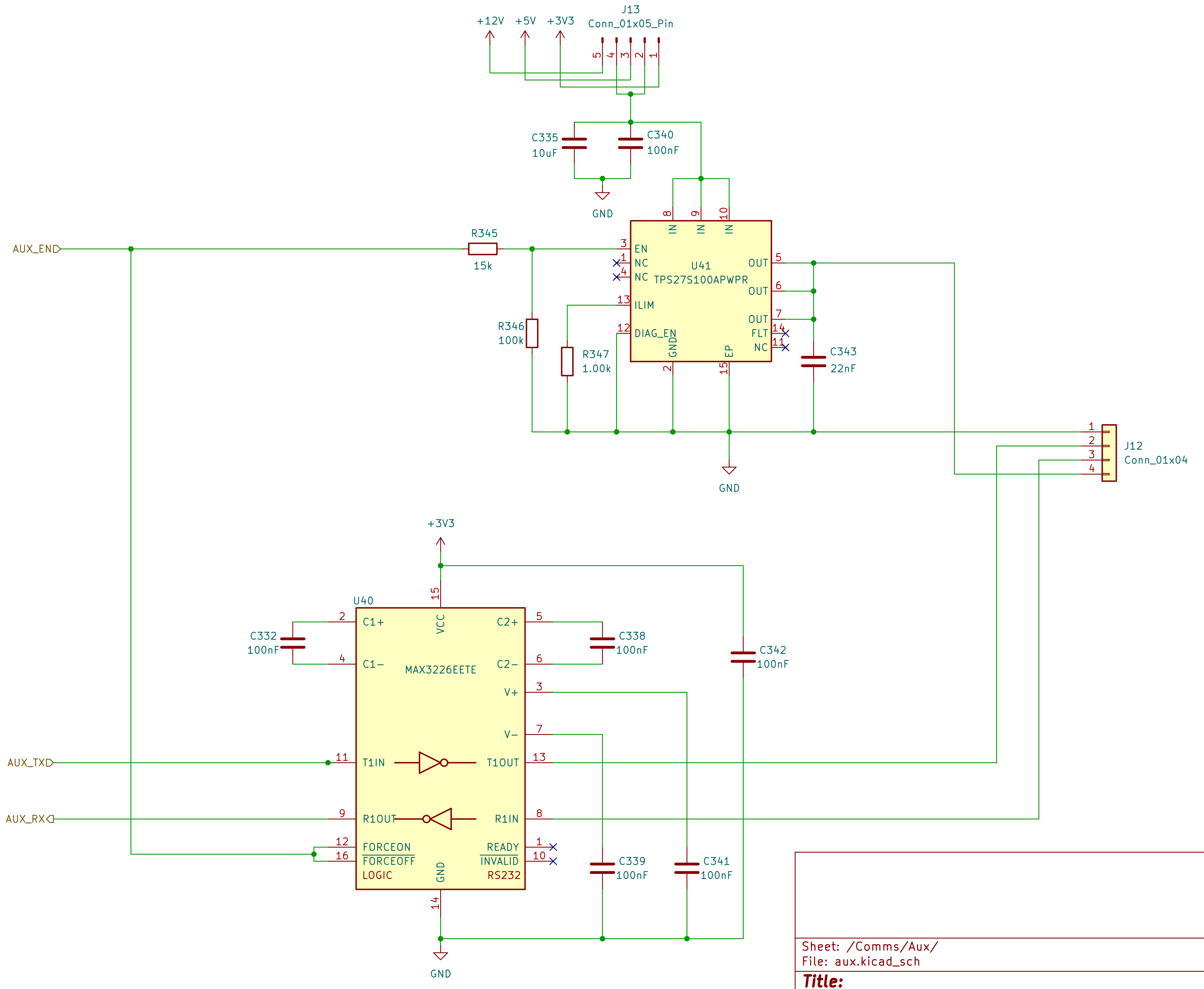
Rev:
KiCad E.D.A. 10.0.4
Id: 23/24



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Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 24/24